

COMPLETE USER GUIDE FOR "UNIT ECONOMICS ASSISTANT"

WHAT IS THIS PROGRAM?

"Unit Economics Assistant" is a script for **Google Sheets** that turns your spreadsheet into a powerful analytical tool. The program helps you:

-  **Calculate key business metrics** (revenue, CAC, LTV, ROMI, etc.)
 -  **Conduct statistical tests** (A/B, A/B/n, t-test, correlation, ANOVA)
 -  **Copy ready-made formulas** for inserting into cells
 -  **Learn** — answers questions about metrics in Russian, English, and German
 -  **Split traffic** — 6 types of user distribution into groups
 -  **Forecast** — ARIMA, Monte Carlo, linear forecast
 -  **Validate data** — normality, outliers, missing values
 -  **Visualize** — charts, dashboards, heatmaps, conversion funnel
 -  **Apply machine learning** — clustering (K-Means)
 -  **Analyze cohorts** — user retention by month
 -  **Use ready-made templates** — marketing, product, financial dashboards
 -  **Work in multiple languages** — Russian, English, German
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HOW TO INSTALL

Step 1: Open Google Sheets

Create a new spreadsheet or open an existing one.

Step 2: Open Apps Script

From the top menu, click:

text

Extensions → Apps Script

Step 3: Paste the code

- Delete all existing code (Ctrl+A → Delete)
- Copy the full program code

- Paste into the editor (Ctrl+V)

Step 4: Save and refresh

- Press **Ctrl+S** → name the project "Unit Economics Assistant"
- Close the editor
- **Refresh the page** (F5)

Step 5: Check the menu

The menu **"Unit Economics"** should appear in the top toolbar.

QUICK START — 3 MAIN FUNCTIONS

What you need	What to enter	What you get
Learn theory	<code>=ANSWER("revenue")</code>	Text with formula and explanation
Get formula	<code>=F("revenue")</code>	<code>=D2*F2*B2</code> (copyable)
Run A/B test	<code>=AB_TEST(250,5000,325,5000)</code>	p-value, uplift, recommendation

1. UNIT ECONOMICS METRICS (15 items)

How to get any metric formula:

Enter in a cell: `=F("metric_name")`

#	Metric	What it shows	What to enter	What you get
1	Revenue	Total company income	<code>=F("revenue")</code>	<code>=D2*F2*B2</code>
2	CAC	Customer acquisition	<code>=F("CAC")</code>	<code>=I2/B2</code>

#	Metric	What it shows	What to enter	What you get
		cost		
3	Conversion	Percentage of visitors who purchased	<code>=F("conversion")</code>	<code>=B2/A2</code>
4	Average Order Value	Average purchase amount	<code>=F("aov")</code>	<code>=SUM(D2:D100)/COUNT(E2:E100)</code>
5	CLTV	Gross profit per customer over lifetime	<code>=F("CLTV")</code>	<code>=(D2-K2)*F2</code>
6	Gross Margin	Profit percentage per euro	<code>=F("margin")</code>	<code>=(D2-K2)/D2</code>
7	Payback Period	Investment return time	<code>=F("payback")</code>	<code>=M2/(L2/12)</code>
8	Retention	Percentage of remaining customers	<code>=F("retention")</code>	<code>=(Q2-R2)/S2</code>
9	Churn	Percentage of lost customers	<code>=F("churn")</code>	<code>=1-T2</code>
10	ARPU	Revenue per user	<code>=F("ARPU")</code>	<code>=N2/COUNTA(range)</code>
11	ROMI	Return on marketing investment	<code>=F("ROMI")</code>	<code>=(N2-O2-I2)/I2</code>
12	COGS	Direct costs per transaction	<code>=F("COGS")</code>	<code>=(D2*G2)+H2</code>
13	APC	Average	<code>=F("APC")</code>	<code>=E2/B2</code>

#	Metric	What it shows	What to enter	What you get
		number of payments per customer		
14	CPC	Cost per click	=F("CPC")	=I2/A2
15	CPA	Cost per action	=F("CPA")	=I2/COUNTIF(J2:J100,"yes")

How to copy a formula:

1. Enter =F("revenue") in cell A1
2. =D2*F2*B2 appears in A1
3. **Click on cell A1**
4. Press **Ctrl+C**
5. Go to the target cell
6. Press **Ctrl+V**

□ Formula copied!

💰 2. FINANCIAL METRICS (10 items)

#	Metric	What it shows	Function	Example	Result
16	ROAS	Advertising return in multiples	=ROAS(revenue, budget)	=ROAS(10000,2000)	5
17	LTV	Customer lifetime value	=LTV(AOV, APC, lifetime)	=LTV(100,2,12)	2400
18	CAC Payback	CAC payback in months	=CAC_PAYBACK(CAC, AOV, COGS)	=CAC_PAYBACK(500,100,30)	7.14
1	NPS	Customer	=NPS(promoters,	=NPS(70,20,10,100)	60

#	Metric	What it shows	Function	Example	Result
9		loyalty index	passives, detractors, total)		
20	Churn MRR	Revenue churn	=CHURN_MRR(lost, initial)	=CHURN_MRR(5000,100000)	5%
21	Gross Profit	Gross profit	=GROSS_PROFIT(revenue, cogs)	=GROSS_PROFIT(10000,3000)	7000
22	Margin %	Profit percentage	=MARGIN(revenue, cogs)	=MARGIN(10000,3000)	70%
23	Burn Rate	Monthly cash burn rate	=BURN_RATE(expenses)	=BURN_RATE(8000)	8000
24	Runway	Months of cash remaining	=RUNWAY(cash, burn_rate)	=RUNWAY(120000,8000)	15
25	GMV	Gross merchandise value	=GMV(orders, AOV)	=GMV(1000,50)	50000

□ 3. STATISTICAL TESTS (15 types)

#	Test	Function	When to use	Example
1	A/B test	=AB_TEST()	Compare 2 groups	=AB_TEST(250,5000,325,5000)
2	A/B/n test	=ABN_TEST()	Compare 3+ groups	=ABN_TEST({{250,5000,"A"}},{275,5000,"B"}})

#	Test	Function	When to use	Example
3	A/A test	<code>=AA_TEST()</code>	Check test correctness	<code>=AA_TEST(250,5000,248,5000)</code>
4	MVT	<code>=MVT_TEST()</code>	Test factor combination s	<code>=MVT_TEST({{...}})</code>
5	Bandit	<code>=BANDIT()</code>	Smart traffic distribution	<code>=BANDIT({{"A",250,4750}})</code>
6	t-test	<code>=T_TEST()</code>	Compare mean values	<code>=T_TEST(A2:A100, B2:B100)</code>
7	Chi-square	<code>=CHI2_TEST())</code>	Test category dependency	<code>=CHI2_TEST({{10,20},{15,25}})</code>
8	ANOVA	<code>=ANOVA()</code>	Compare means in 3+ groups	<code>=ANOVA({{1,2,3},{4,5,6}})</code>
9	Correlation	<code>=CORREL()</code>	Find linear relationship	<code>=CORREL(A2:A100, B2:B100)</code>
10	Mann-Whitney U	<code>=U_TEST()</code>	Nonparametric comparison	<code>=U_TEST(A2:A100, B2:B100)</code>
11	Bayesian A/B	<code>=BAYES_AB()</code>	Probability of test winning	<code>=BAYES_AB(250,4750,325,4675)</code>
12	Normality	<code>=NORMALITY())</code>	Check normality	<code>=NORMALITY(A2:A100)</code>
13	Cohen's d	<code>=COHEN_D()</code>	Effect size	<code>=COHEN_D(A2:A100, B2:B100)</code>
1	Statistical	<code>=STAT_POWER</code>	Probability	<code>=STAT_POWER(0.5, 100)</code>

#	Test	Function	When to use	Example
4	power	<code>()</code>	of detecting effect	
1 5	MDE	<code>=MDE()</code>	Minimum detectable effect	<code>=MDE(1000, 10)</code>

How to interpret p-value in A/B test:

p-value	Meaning	Action
< 0.05	☐ Statistically significant	Implement the test version
0.05 - 0.10	☐ Suggestion	Increase sample size
> 0.10	☐ Not significant	No difference

🎯 4. TRAFFIC SPLITTING (6 types)

Traffic splitting automatically distributes users into groups. The result is **stable** (groups don't change when refreshing the table).

#	Split type	Function	Example	Result
1	Simple (equal)	<code>=SPLIT(ID, 2)</code>	<code>=SPLIT(ROW(), 2)</code>	0 or 1 (50/50)
2	With names	<code>=SPLIT(ID, {"A", "B"})</code>	<code>=SPLIT(ROW(), {"Control", "Test"})</code>	Control or Test
3	Flexible (different shares)	<code>=GROUP_NAME(ID, {"A":50, "B":30, "C":20})</code>	<code>=GROUP_NAME(ROW(), {"C":50, "T1":30, "T2":20})</code>	C(50%), T1(30%), T2(20%)
4	Stratified	<code>=STRATIFIED_SPLIT(range, stratum, {70,30})</code>	Preserves feature proportions	Balanced groups

#	Split type	Function	Example	Result
5	K-Fold	<code>=K_FOLD(range, k)</code>	<code>=K_FOLD(A2:F100, 5)</code>	5 parts for cross-validation
6	Split by date	<code>=SPLIT_BY_DATE(dates, "2024-01-01")</code>	Split before/after date	Data before and after threshold

What can be used as ID:

- Row number: `ROW()`
- User email: A2 (if email is there)
- Any unique number: 12345

5. DATA PROCESSING

#	Function	What it does	Example
1	<code>=CLEAN_DATA(range, method)</code>	Remove outliers, fill missing values	<code>=CLEAN_DATA("A2:A100", "remove_outliers")</code>
2	<code>=NORMALIZE(range)</code>	Scale to 0-1	<code>=NORMALIZE("B2:B100")</code>
3	<code>=STANDARDIZE(range)</code>	Z-score (mean=0, std=1)	<code>=STANDARDIZE("B2:B100")</code>
4	<code>=SAMPLE(range, size)</code>	Random sample from data	<code>=SAMPLE("A2:F1000", 100)</code>
5	<code>=QUARTILES(range)</code>	1st, 2nd, 3rd quartiles	<code>=QUARTILES("A2:A100")</code>

□ 6. CLUSTERING (K-Means)

Function: `=CLUSTER(range, k)`

What it does: Groups data points into k clusters

Example: `=CLUSTER("B2:F1000", 5)`

Result: Splits data into 5 groups, results on a separate sheet "Clusters"

7. COHORT ANALYSIS

Function: `=COHORTS(dates, events, value)`

What it does: Analyzes user retention by month of arrival

Example: `=COHORTS("A2:A1000", "B2:B1000", "purchase")`

Result: Table showing what % of users return each month

8. FORECASTING

#	Function	What it does	Example
1	<code>=ARIMA_FORECAST(range, steps)</code>	ARIMA time series forecast	<code>=ARIMA_FORECAST("B2:B100", 12)</code>
2	<code>=MONTE_CARLO(mean, std, iterations)</code>	Simulate thousands of scenarios	<code>=MONTE_CARLO(100, 15, 10000)</code>
3	<code>=FORECAST(range, periods)</code>	Linear forecast	<code>=FORECAST("B2:B100", 6)</code>

9. VISUALIZATION

9.1 Charts

Function	What it does
<code>=CREATE_CHART("A2:A10", "line", "Sales")</code>	Creates a line chart

Function	What it does
<code>=CHART_AB(250,5000,325,5000)</code>	Creates A/B test chart
<code>=TREND_CHART("A2:A10", "B2:B10", "Revenue")</code>	Trend chart with regression line

9.2 Funnel Diagram

Function: `=FUNNEL(stages, values)`

Example: `=FUNNEL({"Visits","Cart","Purchase"}, {10000,2000,500})`

9.3 Heatmaps

Function	What it does
<code>=HEATMAP("A2:F20")</code>	Colors cells by values
<code>=HEATMAP_CONVERSION(data)</code>	Conversion heatmap by day and hour
<code>=HEATMAP_MATRIX("A1:Z1", "A2:A20", "Metric")</code>	Two-factor matrix

10. READY-MADE DASHBOARDS (via menu)

Click on "**Unit Economics**" menu → select dashboard:

#	Dashboard	What it includes
1	Marketing	ROMI, ROAS, CAC, CPC, CPA, CTR
2	Product	Retention (1,3 months), DAU/MAU, LTV, Churn, NPS
3	Financial	Revenue, COGS, Gross Profit, Margin, Burn Rate, Runway
4	A/B test report	Complete A/B test analysis
5	Funnel	Conversion by stage

#	Dashboard	What it includes
6	Cohort	Retention by cohorts

💡 11. LEARNING — ANSWERS TO QUESTIONS

Function `=ANSWER("question")` — in Russian

Function `=ANSWER_LANG("question", "english")` — in English

Function `=ANSWER_LANG("question", "german")` — in German

What to ask

What you learn

`=ANSWER("revenue")`

Formula: Revenue = AOV × APC × B

`=ANSWER("CAC")`

CAC = Marketing budget ÷ B

`=ANSWER("conversion")`

CR = B ÷ UA × 100%

`=ANSWER("CLTV")`

CLTV = (AOV – COGS) × APC

`=ANSWER("A/B test")`

Step-by-step instructions

`=ANSWER("splitting")`

How to split traffic into groups

`=ANSWER("chart")`

How to create a chart

`=ANSWER("dashboard")`

How to create a dashboard

`=ANSWER("heatmap")`

How to make a heatmap

`=ANSWER("cohorts")`

How to do cohort analysis

`=ANSWER("clustering")`

How to group data

12. CALCULATION FUNCTIONS

#	Function	What it does	Example	Result
1	<code>=SAMPLE_SIZE(base, uplift)</code>	How many users needed	<code>=SAMPLE_SIZE(10, 20)</code>	3700 per group
2	<code>=UPLIFT(control, test)</code>	Relative change	<code>=UPLIFT(100, 120)</code>	+20%
3	<code>=SCORE("name", impact, confidence, ease)</code>	Hypothesis priority (ICE)	<code>=SCORE("New design", 8, 7, 9)</code>	ICE = 8.0
4	<code>=COHEN_D(group1, group2)</code>	Effect size	<code>=COHEN_D(A2:A100, B2:B100)</code>	0.8 → large effect
5	<code>=STAT_POWER(0.5, 100)</code>	Statistical power	<code>=STAT_POWER(0.5, 100)</code>	85% → sufficient
6	<code>=MDE(1000, 10)</code>	Minimum detectable effect	<code>=MDE(1000, 10)</code>	5.6%

13. MULTILANGUAGE SUPPORT

Function	Language	Example	Result
<code>=F_LANG("revenue", "english")</code>	English	<code>=F_LANG("revenue", "english")</code>	<code>=D2*F2*B2</code>
<code>=F_LANG("umsatz", "german")</code>	German	<code>=F_LANG("umsatz", "german")</code>	<code>=D2*F2*B2</code>
<code>=ANSWER_LANG("revenue", "english")</code>	English	<code>=ANSWER_LANG("revenue", "english")</code>	Theory in English
<code>=ANSWER_LANG("umsatz", "german")</code>	German	<code>=ANSWER_LANG("umsatz", "german")</code>	Theorie auf Deutsch

14. MENU IN THE SPREADSHEET

After installation, the **"Unit Economics"** menu appears in the top toolbar:

text

Unit Economics

- |— Help
 - |— Marketing Dashboard
 - |— Product Dashboard
 - |— Financial Dashboard
 - |— A/B Test Report
 - |— Insert Examples
-

CHEAT SHEET (QUICK ACCESS)

What you need

What to enter

Learn theory

`=ANSWER("revenue")`

Get formula

`=F("revenue")`

A/B test

`=AB_TEST(250,5000,325,5000)`

Sample size

`=SAMPLE_SIZE(10,20)`

Correlation

`=CORREL(A2:A100, B2:B100)`

Hypothesis score

`=SCORE("Idea",8,7,9)`

Split 50/50

`=SPLIT(ROW(), 2)`

Split with names

`=SPLIT(ROW(), {"A","B"})`

Forecast

`=FORECAST("B2:B100", 12)`

Funnel

`=FUNNEL({"A","B","C"},{1000,500,100})`

Heatmap

`=HEATMAP("A2:F20")`

What you need

What to enter

Theory in English

`=ANSWER_LANG("revenue","english")`

💡 TIPS FOR USE

1. **Use `ROW()` instead of ID** — row number as a unique identifier
 2. **Formulas from `=F()` are copyable** — click on the cell → Ctrl+C → Ctrl+V
 3. **Splitting is stable** — groups don't change when refreshing the table
 4. **A/B test gives p-value** — if < 0.05 , statistically significant
 5. **For large datasets, use arrays** — functions work with ranges
 6. **Combine functions** — you can use one function's result as another's argument
 7. **Use the menu for dashboards** — no need to enter formulas
-

❏ FREQUENTLY ASKED QUESTIONS

Q1: How to copy a formula from `=F()`?

A: Click on the cell with the result → Ctrl+C → go to target cell → Ctrl+V

Q2: Why does `=ANSWER()` return "Question not recognized"?

A: Use exact words: revenue, CAC, conversion, CLTV, AOV, margin

Q3: How to know if an A/B test is significant?

A: $p\text{-value} < 0.05$ → significant

Q4: How many users are needed for a test?

A: `=SAMPLE_SIZE(10, 20)` → 3700 per group

Q5: How to split users into groups?

A: `=SPLIT(ROW(), {"Control", "Test"})`

Q6: How to create a dashboard?

A: Click on the "Unit Economics" menu → desired dashboard

Q7: What is p-value in simple terms?

A: The probability of making a mistake by saying "there is an effect" when there isn't

Q8: How often should dashboard data be updated?

A: The dashboard is created once; data updates automatically

FINAL STATISTICS

Category	Quantity
Unit economics metrics	15
Financial metrics	10
Statistical tests	15
Traffic splitting types	6
Data processing	5
Clustering	1
Cohort analysis	1
Forecasting	3
Visualization	7
Ready-made dashboards	6
Learning (answers)	15+
Calculation functions	6

Category	Quantity
Multilanguage	3 languages
TOTAL FUNCTIONS	~90+